

0/3 6.2.2.

$$\square X_n = 2^n + 1$$

$$X_1 = 2^1 + 1 = 2 + 1 = 3$$

$$X_2 = 2^2 + 1 = 4 + 1 = 5$$

$$X_3 = 2^3 + 1 = 8 + 1 = 9$$

$$X_4 = 2^4 + 1 = 16 + 1 = 17$$

6.2.3

$$\square X_n = n^2 + 2n + 3$$

$$X_1 = 1^2 + 2 \cdot 1 + 3 = 1 + 2 + 3 = 6$$

$$X_2 = 2^2 + 2 \cdot 2 + 3 = 4 + 4 + 3 = 11$$

$$X_3 = 3^2 + 2 \cdot 3 + 3 = 9 + 6 + 3 = 18$$

$$X_4 = 4^2 + 2 \cdot 4 + 3 = 16 + 8 + 3 = 27$$

6.2.4.

$$\square X_n = (-1)^n + 1$$

$$X_1 = (-1)^1 + 1 = -1 + 1 = 0$$

$$X_2 = (-1)^2 + 1 = 1 + 1 = 2$$

$$X_3 = (-1)^3 + 1 = -1 + 1 = 0$$

$$X_4 = (-1)^4 + 1 = 1 + 1 = 2$$

6.2.5

$$\square \quad X_n = \frac{n+1}{n^2}$$

$$X_1 = \frac{1+1}{1^2} = \frac{2}{1} = 2$$

$$X_3 = \frac{3+1}{3^2} = \frac{4}{9}$$

$$X_2 = \frac{2+1}{2^2} = \frac{3}{4}$$

$$X_4 = \frac{4+1}{4^2} = \frac{5}{16}$$

6.2.6

$$\square \quad X_n = \sin \frac{\pi n}{2}$$

$$X_1 = \sin \frac{\pi}{2} = 1$$

$$X_2 = \sin \frac{2\pi}{2} = \sin \pi = 0$$

$$X_3 = \sin \frac{3\pi}{2} = -1$$

$$X_4 = \sin \frac{4\pi}{2} = \sin 2\pi = 0$$

6.2.7

$$\square \quad X_1 = -1 \quad X_n = -n \cdot X_{n-1}$$

$$X_2 = -2 \cdot (-1) = 2$$

$$X_3 = -3 \cdot (2) = -6$$

$$X_4 = -4 \cdot (-6) = 24$$

6.2.13

$$X_n = (-1)^n$$

$$\square \quad \{X_n\} = \{-1, 1, -1, 1, -1, \dots\}$$

$$\min(X_i) = -1 \quad \max(X_i) = 1 \Rightarrow \text{Огранич. сверху}$$

и снизу.

6.2.14

$$x_n = n^3 + 2n$$

$$\square \{x_n\} = \{3, 12, 33, 92, 135, \dots\}$$

$\min(x_i) = 3 \Rightarrow$ ограничена снизу 3

6.2.15

$$x_n = -\ln n$$

$$\square \{x_n\} = \{0; -\ln 2; -\ln 3; -2\ln(2) \dots\}$$

$\max(x_i) = 0 \Rightarrow$ огранич сверху. 0

6.2.16

$$x_n = \frac{n+1}{n}$$

$$\square \{x_n\} = \{2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \dots\}$$

$\max(x_i) = 2 \Rightarrow$ огранич. сверху 2

$x_n \rightarrow 0, \forall n \in \mathbb{N} \Rightarrow$ огранич снизу 0

6.2.17

$$\square x_n = (-1)^n \cdot n$$

$$\{x_n\} = \{-1, 2, -3, 4, \dots\}$$

$$\forall M > 0 \exists n : |x_n| = n > M$$

$\Rightarrow$ не огранич

6.2.18

$$\square \quad X_n = \begin{cases} 1, & \text{при } n=2k \\ \sqrt{n}, & \text{при } n=2k+1 \end{cases}$$

$$\{X_n\} = 1, 1, \sqrt{3}, 1, \sqrt{5} \dots$$

$\min_{i \in \mathbb{N}} (X_i) = 1 \Rightarrow$ ограни снизу 1 $\blacksquare$

6.2.20

$$X_n = n - \frac{1}{n}$$

$$\square \quad \text{ ~~$X_n = (n+1) - \frac{1}{n+1}$~~ }$$

$$X_n = \{n - \frac{1}{n}\} = \{1, \frac{3}{2}, \frac{8}{3}, \frac{15}{4}, \frac{24}{5} \dots\}$$

Послед строго возраст $\Rightarrow$

$\Rightarrow$ строго монотонная $\blacksquare$

6.2.21

$$\square \quad X_n = \cos \frac{\pi n}{2}$$

$$X_n = \{\cos \frac{\pi n}{2}\} = \{0, -1, 0, 1, 0 \dots\}$$

$\max(X_i) = 1, \min(X_i) = -1 \Rightarrow$ ограни

не монотон. $\blacksquare$

6.2.22

$$\square X_n = -\frac{n^2+1}{n^2}$$

$$X_n = \left\{ -2, -\frac{5}{4}, -\frac{10}{9}, -\frac{17}{16}, -\frac{26}{25} \dots \right\}$$

$$\min(x_i) = -2 \quad X_n \rightarrow 0 \Rightarrow \text{огранич}$$

посл. $\uparrow \Rightarrow$ строго монотонная

6.2.23

$$\square X_n = -\sqrt{n}$$

$$X_n = \left\{ -1, -\sqrt{2}, -\sqrt{3}, -2, -\sqrt{5} \dots \right\}$$

$$\max(x_i) = -1 \Rightarrow \text{огранич сверху}$$

строго $\downarrow \Rightarrow$ строго монотонная

6.2.24

$$\square X_n = \pi, \pi, \pi \dots$$

послед. не $\uparrow$ и не $\downarrow \Rightarrow$ монотонная

$$\max(x_i) = \min(x_i) = \pi \Rightarrow \text{ограничен.}$$

6.2.26

$$\square \quad x_n = (-1)^n \quad y_n = (-2)^n$$

$$\{x_n + y_n\} = \{(-1)^n + (-2)^n\} =$$

$$= \{-3, 5, -9, 17, -33, \dots\}$$

$$\{x_n - y_n\} = \{(-1)^n - (-2)^n\} =$$

$$= \{1, -3, 7, -15, 31, \dots\}$$

$$\{x_n \cdot y_n\} = \{(-1)^n \cdot (-2)^n\} =$$

$$= \{2, 4, 8, 16, 32, \dots\}$$

$$\left\{\frac{x_n}{y_n}\right\} = \left\{\frac{(-1)^n}{(-2)^n}\right\} = \left\{0.5, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots\right\}$$

6.2.27

$$\square \quad x_n = n^2 + 1 \quad y_n = n$$

$$\{x_n + y_n\} = \{n^2 + n + 1\} = \{3, 7, 13, 21, 31, \dots\}$$

$$\{x_n - y_n\} = \{n^2 - n + 1\} = \{1, 3, 7, 13, 21, \dots\}$$

$$\{x_n \cdot y_n\} = \{n(n^2 + 1)\} = \{2, 10, 30, 68, 130, \dots\}$$

$$\left\{\frac{x_n}{y_n}\right\} = \left\{\frac{n^2 + 1}{n}\right\} = \left\{2, \frac{5}{2}, \frac{10}{3}, \frac{17}{4}, \frac{26}{5}, \dots\right\}$$

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$$\square X_n = (\sqrt{2})^n \quad y_n = 1 \quad \alpha = \sqrt{2} \quad \beta = -5$$

$$\alpha X_n + \beta y_n = \{ \sqrt{2} \cdot (\sqrt{2})^n + (-5) \cdot 1 \} =$$

$$= \{ (\sqrt{2})^{n+1} - 5 \} = \{ \sqrt{2} - 5, -3, 2\sqrt{2} - 5, -1, 4\sqrt{2} - 5 \}$$

